Data Science

Project Report

Predictive Modeling and Micro-Climate Mitigation of Urban Heat Islands in Karachi Using GEE and Machine Learning

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1. Abstract

Urban Heat Islands (UHIs) are a growing environmental concern in rapidly urbanizing cities such as Karachi, where increasing built-up areas and decreasing vegetation contribute to rising land surface temperatures. This study utilizes satellite-derived geospatial data to analyze the relationship between environmental indices and Land Surface Temperature (LST). A dataset containing vegetation (NDVI), built-up (NDBI), water (NDWI), and surface reflectance (Albedo) indices was analyzed for the years 2015–2018.

Four machine learning models—Random Forest, XGBoost, Support Vector Regression (SVR), and Multi-Layer Perceptron (MLP)—were developed to predict LST. Among these, XGBoost achieved the best performance with an R^2 score of 0.869. Additionally, a mitigation simulation was conducted by increasing vegetation index values to evaluate potential cooling effects, demonstrating a measurable reduction in surface temperature.

2. Introduction

Urbanization significantly alters land surface characteristics, leading to the formation of Urban Heat Islands (UHIs), where urban regions experience higher temperatures than surrounding rural areas. In cities like Karachi, this phenomenon is intensified due to dense construction, reduced vegetation, and limited water bodies.

Traditional approaches to studying UHIs rely on field measurements, which are often limited in spatial and temporal coverage. With advancements in remote sensing and machine learning, satellite data can now be used to model and predict urban temperature dynamics.

This study aims to:

- Analyze spatial and environmental factors contributing to UHI
- Predict Land Surface Temperature using machine learning models
- Evaluate the impact of vegetation changes through simulation

3. Dataset Description

The dataset used in this study was derived from satellite imagery processed through geospatial tools including Google Earth Engine. It contains approximately 40,000 spatial observations collected from Karachi.

Features:

- NDVI (Vegetation Index)
- NDBI (Built-up Index)
- NDWI (Water Index)
- Albedo (Surface reflectivity)
- Latitude & Longitude
- Year (2018–2025)

Target Variable:

- LST (Land Surface Temperature)

3.1 Data Acquisition via Google Earth Engine (GEE)

The primary dataset for this study was dynamically extracted using the Google Earth Engine (GEE) cloud computing platform. Instead of relying on static secondary data, a custom JavaScript algorithm was developed to retrieve and process high-resolution satellite imagery from the **Landsat 8 (Collection 2, Tier 1, Level 2)** dataset.

The data extraction process involved several key technical steps:

- **Region of Interest (ROI) Demarcation:** To ensure data accuracy and eliminate irrelevant oceanic pixels, a geographical bounding box of Karachi was intersected with the official Pakistan administrative boundary (USDOS/LSIB_SIMPLE/2017). This created a bulletproof land-only mask for the city.
- **Cloud Masking:** A strict cloud-masking function was applied utilizing the QA_PIXEL band. By applying bitwise operations to mask out cloud and cloud shadow bits, the algorithm ensured that only clear, cloud-free surface pixels were used for analysis.
- **Feature Engineering & Indices Calculation:** Environmental indices were calculated directly on the cloud infrastructure using specific Landsat 8 spectral bands:
    - **NDVI, NDBI, and NDWI** were calculated using normalized difference functions between the Near-Infrared (NIR), Red, Green, and Shortwave-Infrared (SWIR) bands.
    - **Albedo (Surface Reflectivity)** was estimated using a broad-band empirical formula.
    - **Land Surface Temperature (LST)** was derived from the Thermal Band (ST_B10) and converted from Kelvin to Celsius.
- **Spatio-Temporal Sampling:** To capture seasonality and micro-climate variations, the algorithm was designed to extract monthly median composites from **2018 to 2025**. A stratified random sampling technique was used to extract 500 spatial points (pixels) per month, yielding 6,000 data points per year.
- **Coordinate Extraction:** To map the data back to specific districts (e.g., Saddar, Korangi, Malir) during the EDA phase, precise Latitude and Longitude coordinates were appended to every single pixel observation before exporting the final CSV files to Google Drive.

4. Methodology

4.1 Data Preprocessing:

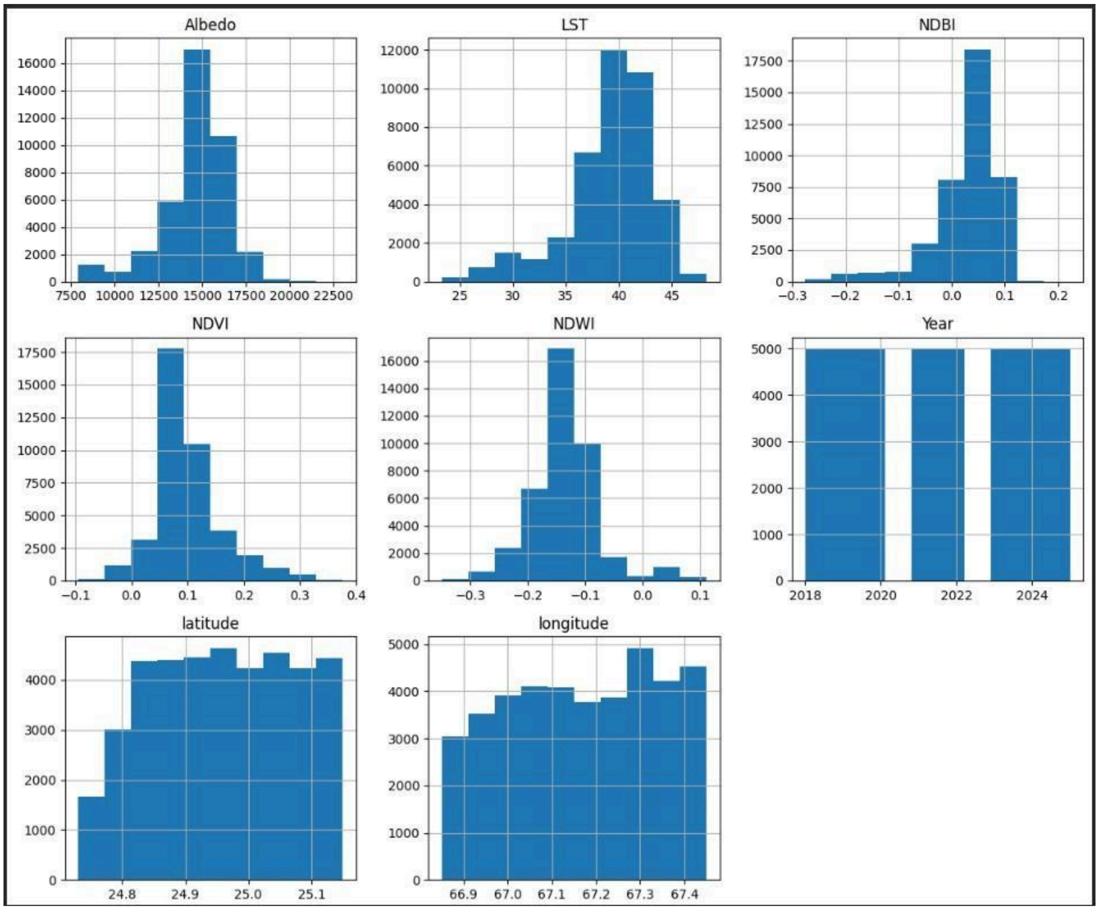
- Removal of irrelevant columns (system:index, .geo)
- Outlier treatment using IQR-based capping
- Feature scaling using StandardScaler
- Train-test split (80:20)
- Performed feature engineering

4.2 Exploratory Data Analysis (EDA):

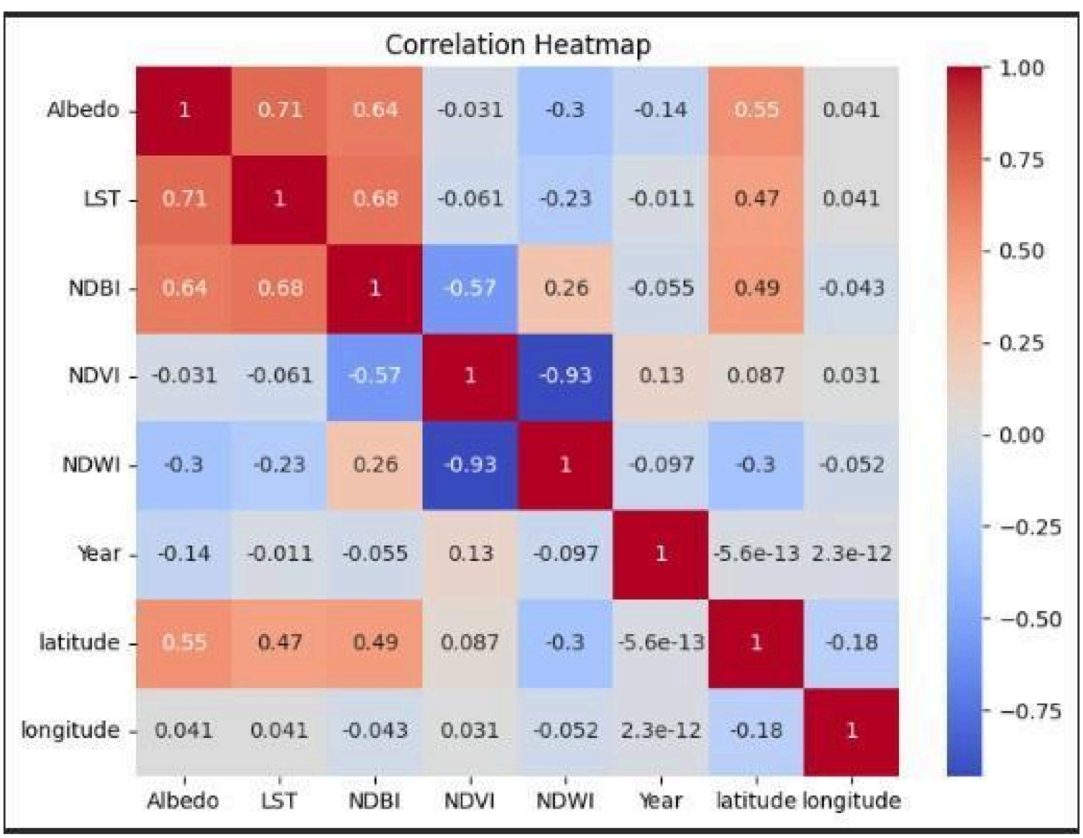
EDA was performed to understand relationships between variables:

- Correlation analysis showed:
- Positive correlation between LST and NDBI
- Negative correlation between LST and NDVI and NDWI
- Spatial analysis identified heat concentration in highly urbanized regions ● Distribution analysis confirmed variability across environmental indices

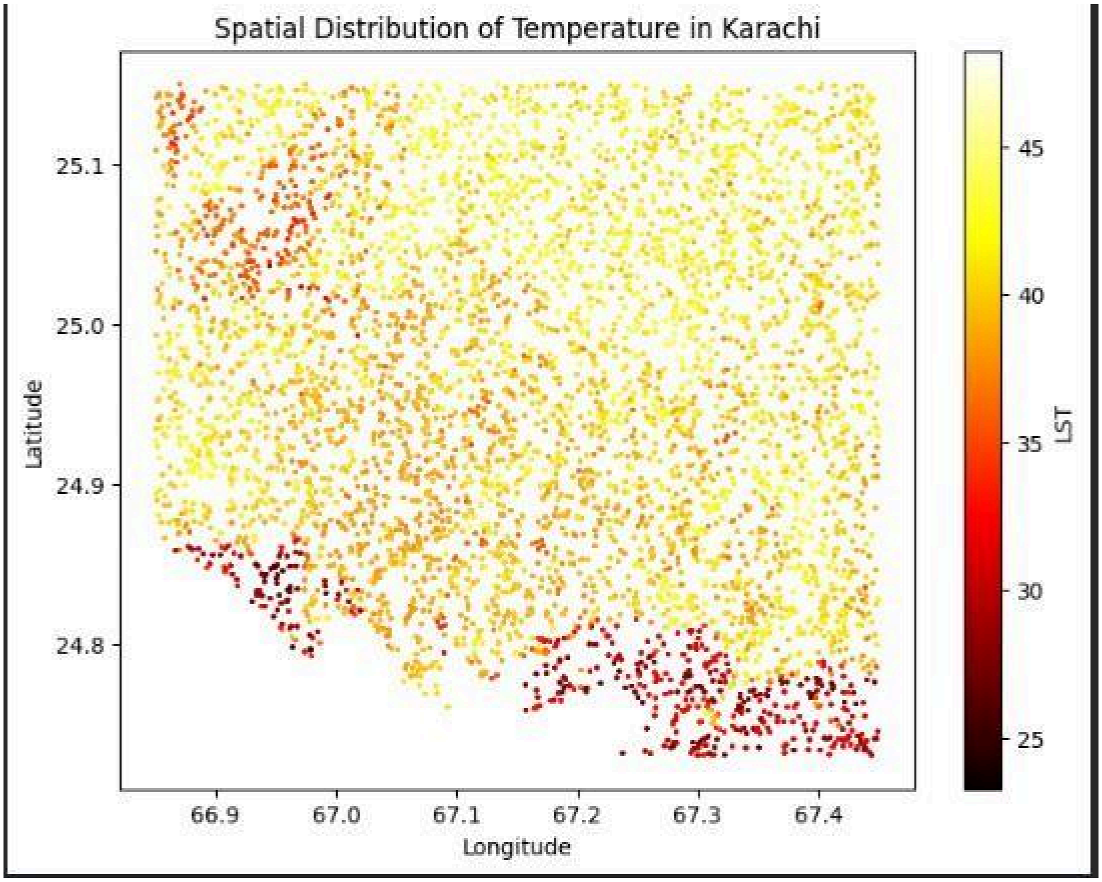
Distribution Plot:



Correlation Heatmap:

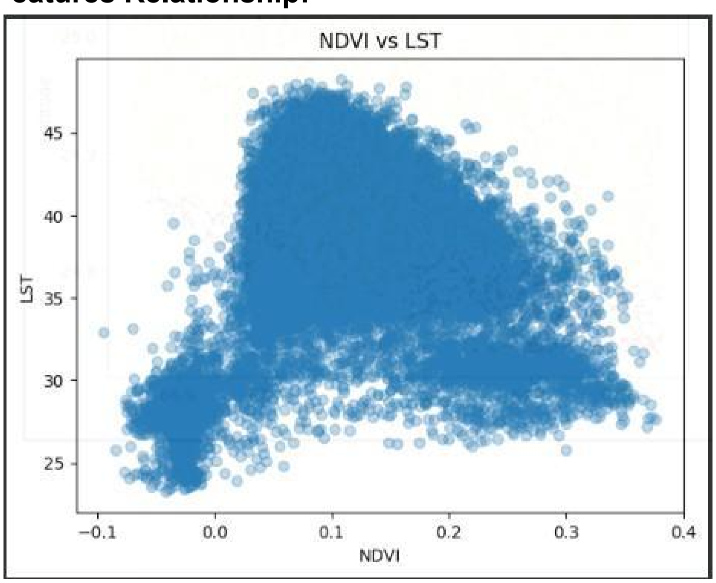


Spatial Distribution:



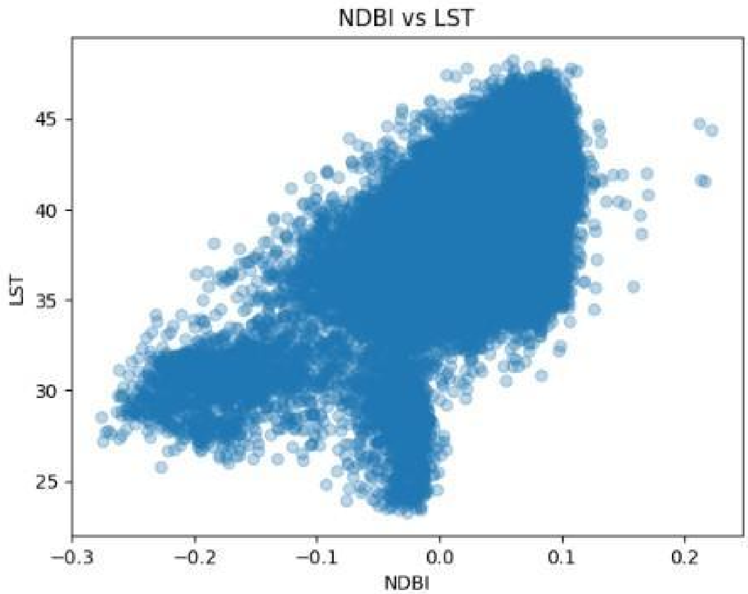
We can infer that coastal areas has observed less temperature then the densely populated with concrete buildings area like(Saddar, Korangi, Lyari)

Features Relationship:

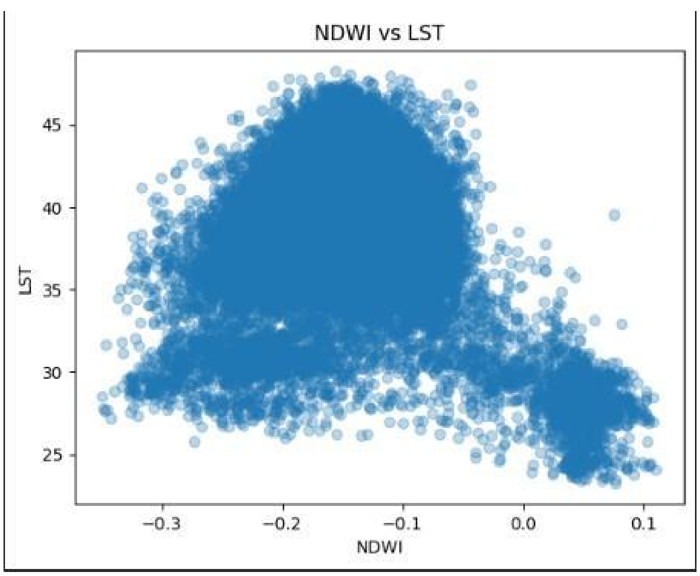


We can infer that range -0.1 till 0 has observed less LST because of the waterbodies,

From range 0 till 0.1 has increased highest LST (low vegetation -> High LST), then after that increase in vegetation resulted in decrease in LST

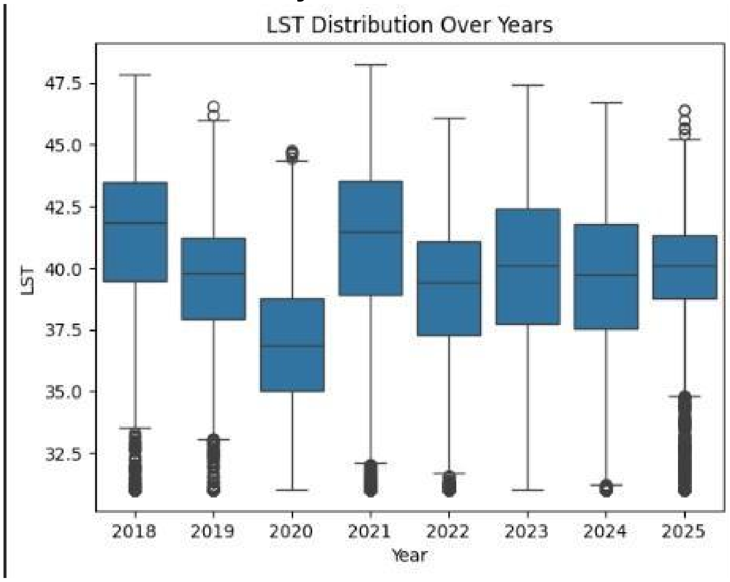


We can infer that NDBI has positive correlation with LST, from -0.3 to 0 are the water bodies which does not contain buildings e.t.c, from 0 till 0.2 we observed that there is always increase in LST with increase in NDBI



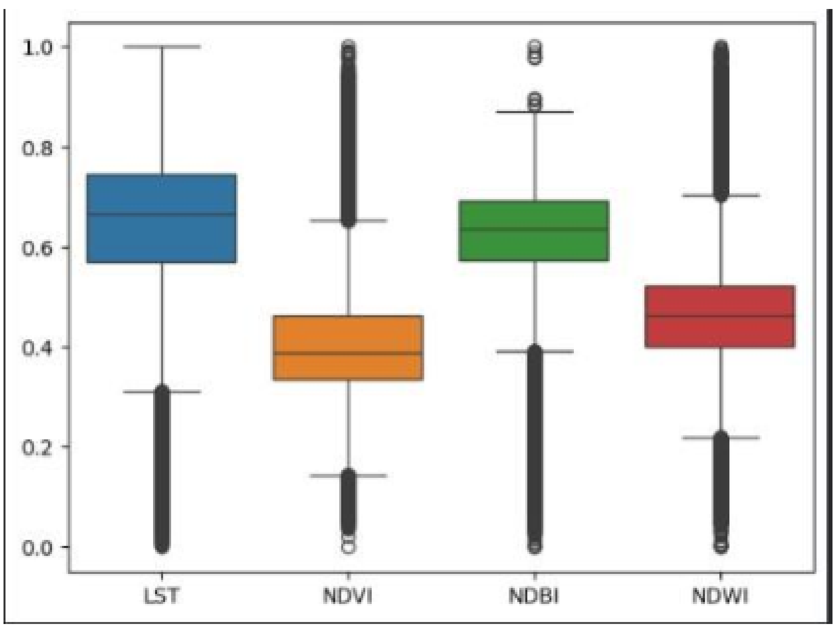
We can infer that NDWI has negative correlation with LST, from -0.3 to 0 we can see the temperature is high because of built-up areas or barren soil land, from 0 till 0,1 we can see that there is decline in LST with the increase in NDWI.

LST Distribution over years:



We can observe that year 2020 was the coldest year with average of 37.5C Temperature, then there is slight increase in temperature patterns

Outliers:




```
...  Albedo       2487
     LST          2578
     NDBI         2163
     NDVI         3036
     NDWI         2583
     Year            0
     latitude        0
     longitude       0
     dtype: int64
```

These were the amount of outliers in the data, to handle these outliers properly we use Outlier Capping using the IQR method, this ensures that important data was not deleted but just adjusted to the lower and upper bound, which ultimately reduces the noise.

4.3 Model Building:

Four regression models were implemented:

- Random Forest Regressor
- XGBoost Regressor
- Support Vector Regression (SVR)
- Multi-Layer Perceptron (MLP)

With these performance:

Model	RMSE ↓	MAE ↓	R^2 ↑
XGBoost	**1.340**	**1.019**	**0.863**
Random Forest	1.374	1.019	0.856
MLP (Neural Net)	1.525	1.178	0.823
SVR	1.803	1.338	0.752

We observed that :

- XGBoost (Top Performer): Achieved the highest accuracy (R^2 = 0.863) and lowest error (RMSE = 1.34). It proved most effective at generalizing the complex, non-linear relationships between urban indices and LST.
- Random Forest (Runner-up): Delivered strong, robust results (R^2 = 0.856) with an (RMSE = 1.37). It serves as a highly reliable baseline for spatial variance.
- MLP (Moderate): Showed competitive performance but was unable to surpass the ensemble boosting power of XGBoost.
- SVR (Underperformer): Recorded the lowest accuracy (R^2 = 0.75). Its performance was hindered by the high dimensionality of the geospatial data and the computational scale of the dataset.

Conclusion: XGBoost is the optimal choice for estimating LST in Karachi, though the narrow margin between it and Random Forest suggests both ensemble methods are well-suited for this climate data.

4.4 HyperParameter Tuning:

We applied Hyperparameter Tuning using a technique called **Randomized Search Cross-Validation**. Which gave us these results (new values for XGBoost Parameters)

```
Best Params: {'subsample': 0.8, 'n_estimators': 300, 'max_depth': 6, 'learning_rate': 0.1}
```

Before Tuning:

```
XGBoost Performance:
RMSE: 1.3402412638468015
MAE: 1.0189715512670912
R2: 0.8631351223412558
```

XGBoost Improved Performance:

```
XGBoost Performance:
RMSE: 1.3111844966868267
MAE: 0.9970999816096002
R2: 0.8690053211266899
```

To improve model performance, hyperparameter tuning was conducted using Randomized Search Cross-Validation. The optimized parameters were applied to retrain the XGBoost model, resulting in a marginal improvement in evaluation metrics compared to the initial model. This indicates better generalization and robustness after tuning.

5. Mitigation Simulation:

A mitigation scenario was created by increasing NDVI values by 20% to simulate urban greening. The trained XGBoost model was used to predict resulting temperature changes. And we observed that :

```
Average Temperature Reduction: 0.026113499
```

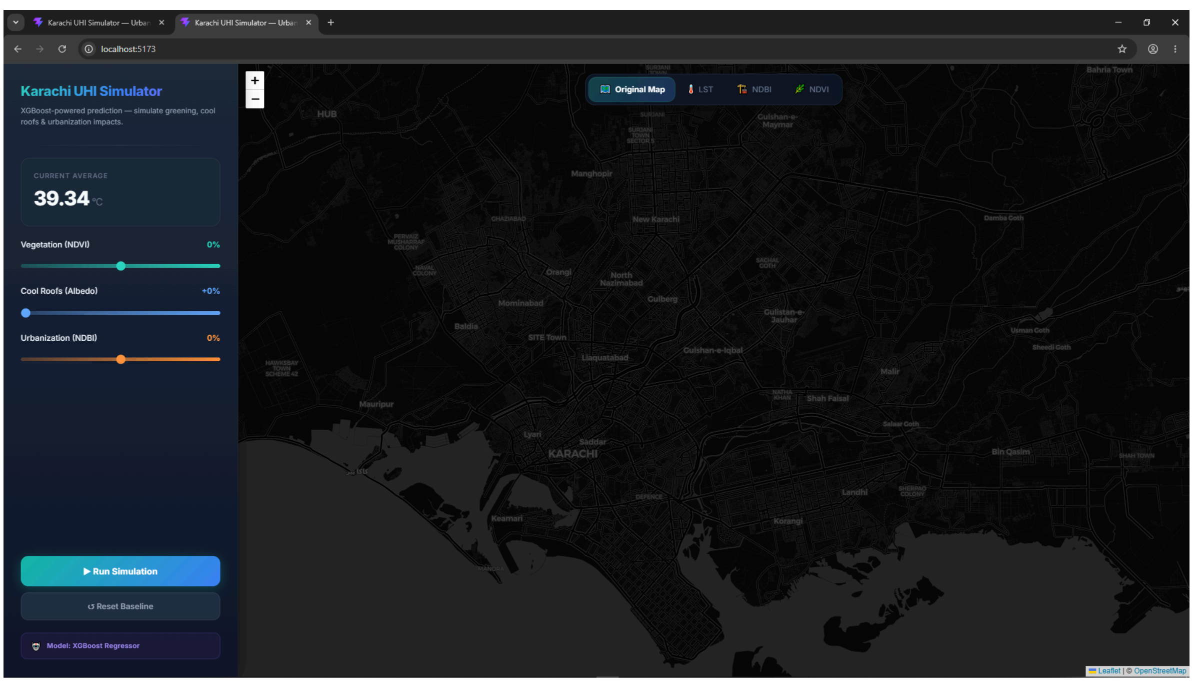
Which proves that A small increase in vegetation produces a small but measurable cooling effect

We implemented a full-stack predictive simulation pipeline to visualize and test our model. The backend utilizes FastAPI to handle spatial data and run inference via our optimized XGBoost model. The frontend is a modern web application built with React and Vite, featuring an interactive Leaflet map that renders multiple geospatial indices (LST, NDBI, NDVI) and allows users to simulate various climate mitigation scenarios.

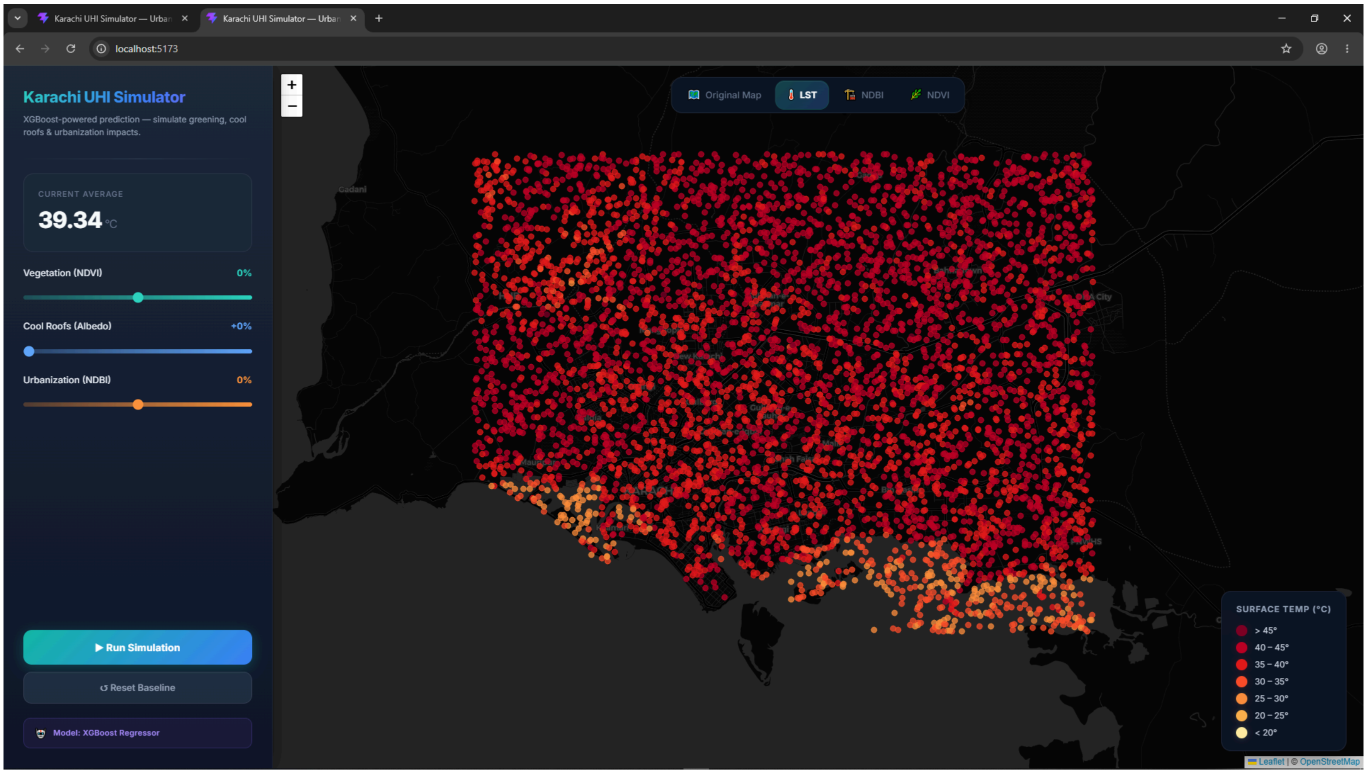
The Simulation Setup:

- The Model: XGBoost (The core predictive engine)
- The Backend: Python & FastAPI (Handling API requests & model inference)
- The Frontend Framework: React & Vite (For a fast, modern UI)
- The Mapping Library: Leaflet (For rendering the geographical data points)
- The Purpose: Simulating environmental interventions (NDVI, NDBI, Albedo)

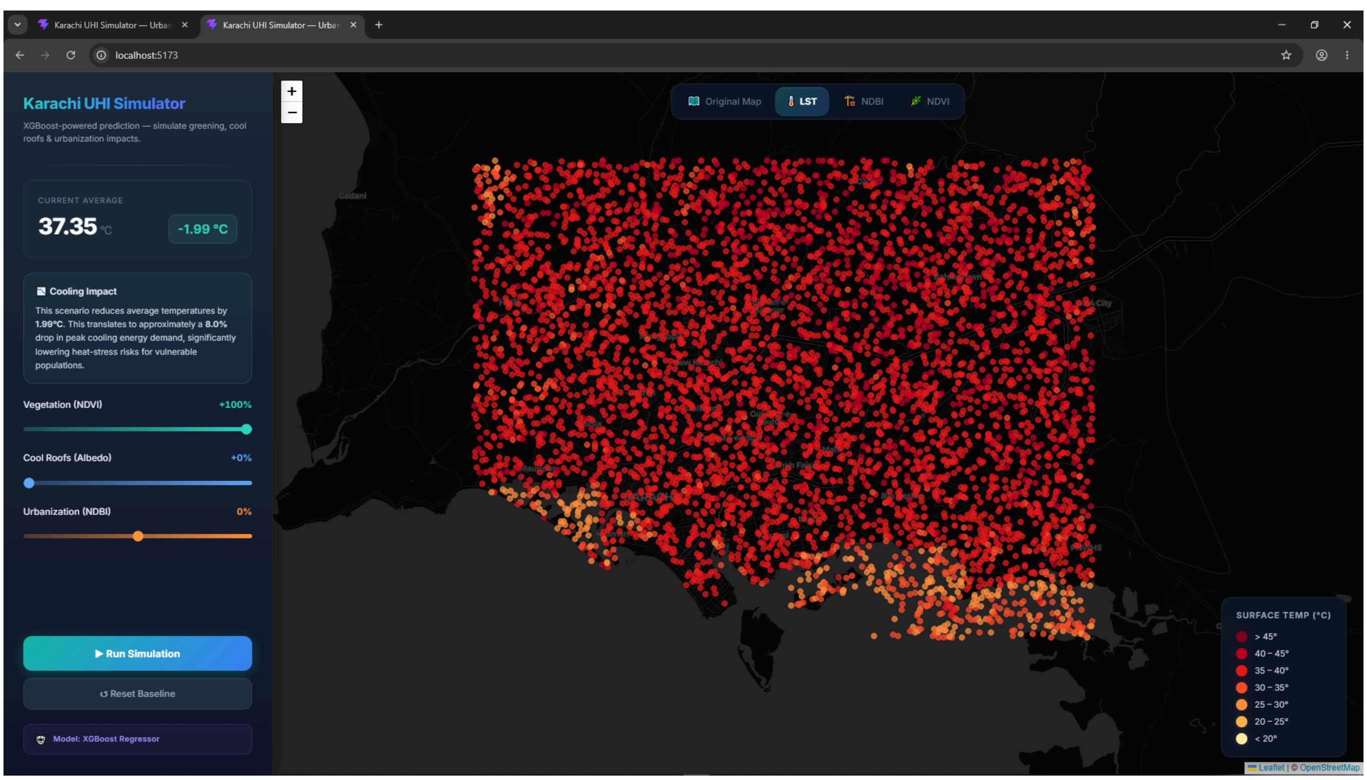
Map View:



LST View:



Simulated Scenario:



6. Discussion:

The results confirm that urban infrastructure plays a significant role in temperature variation. Built-up areas significantly increase surface temperature, while vegetation and water bodies have a cooling effect.

The superior performance of XGBoost and Random Forest indicates that ensemble learning methods are more suitable for capturing non-linear relationships in geospatial environmental data compared to SVR and MLP.

The mitigation simulation demonstrates that vegetation increase alone produces limited cooling, suggesting that comprehensive urban planning strategies are required to effectively reduce urban heat.

7. Conclusion:

This study successfully demonstrates the application of machine learning in analyzing and predicting Urban Heat Island effects in Karachi. XGBoost emerged as the most effective model with high predictive accuracy. The simulation experiment provided valuable insight into the potential impact of urban greening, highlighting the importance of multi-factor urban planning strategies.

This project demonstrates how remote sensing and machine learning can be combined to support climate-aware urban development policies.

8. Future Work

Integration of real-time satellite data streams
Inclusion of humidity and wind speed variables
Deep learning models with spatial convolution (CNNs)
Expansion to other cities for comparative analysis